

Hidden Negative Directions in Surcomplex Linear Algebra

Prime-denominator approximation barriers
and a two-scale matrix-measure criterion

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Abstract

Let $K_\Gamma = \mathbb{C}((t^\Gamma))$, with Γ a nonzero divisible ordered abelian group, and let $\mathcal{P}_\Gamma$ consist of those Hahn series whose support lies in a finitely generated additive subgroup of Γ . This is an algebraically closed subfield containing every monomial. Nevertheless, for every $r \geq 1$ we construct an explicit Hermitian matrix of size $r + 1$ whose quadratic form is strictly positive at every nonzero vector with coordinates in $\mathcal{P}_\Gamma$, and even in its valuation closure, but which has inertia $(1, r, 0)$ over K_Γ ; it is therefore not positive semidefinite. Throughout, “positive semidefinite” means nonnegative at *every* vector over K_Γ ; statements about a restricted set of test vectors always name that set.

The construction rests on prime-denominator tails. For $h = \sum_{p \text{ prime}} t^{\delta(1-1/p)}$, every finite-lattice-supported approximant differs from h at valuation below δ . Matching its first N terms by an element algebraic over $\mathbb{C}(t^\delta)$ requires, and can be achieved with, degree exactly $\prod_{j=1}^N p_j$. Almost-disjoint prime sets yield continuum many algebraically independent tails and a uniform linear-combination separation estimate. These results transport to actual surreal normal forms, including $t = \omega^{-1}$.

A second theorem characterizes positivity of a two-scale matrix-valued measure $M_0 + \varepsilon M_1$ by ordinary scalar null ideals together with a vector-valued absolute-continuity condition. The latter detects the coupling into a residual kernel that scalar positivity misses. It is the two-scale case of the matrix-valued null-ideal criterion that the collection’s Hahn–Herglotz report excludes from its scope. All assertions proposed as contributions have written proofs. The support-field construction, character automorphisms, finite spectral algebra, and elementary density obstruction are explicitly treated as background, not new discoveries.

Status. An AI-assisted research draft: a candidate-original theorem package, not refereed. None of its theorems is Lean-verified; the collection’s Lean ledger covers the scalar two-scale criterion of another report, which Corollary 10.3 recovers as the case $d = 1$. No named published conjecture is claimed settled. The repository comparison is pinned and targeted, not exhaustive (Appendix A). The accompanying finite checks are diagnostics, not proofs of the infinite statements. Every label carries the prefix `hnd:`.

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1 The problem and the contribution

Positivity of a Hermitian matrix is a universal assertion about vectors:

$$A \geq 0 \iff z^* A z \geq 0 \text{ for every } z. \quad (1)$$

Over the ordinary complex numbers, testing a dense collection of vectors is sufficient, by continuity. Over a Hahn field, an algebraically large collection need not be dense. Even an algebraically closed subfield containing every monomial can miss an entire negative cone. This paper makes that failure explicit and quantifies the obstruction.

The question is not whether finite-dimensional spectral theory survives an ordered field extension. It does in the real-closed setting, and the repository already develops that subject [8]. The question is instead how much of a Hahn vector must be available before quadratic-form testing detects the actual sign. Finite exponent groups, algebraic extensions, and valuation limits are three natural sources of probes. Our first theorem defeats all three simultaneously.

1.1 A precise headline result

We use increasing Hahn exponents, so t^γ is infinitesimal when $\gamma > 0$. Write $\mathcal{P}_\Gamma$ for the union of the fields $\mathbb{C}((t^H))$ over finitely generated additive subgroups $H \leq \Gamma$. The term *finite-lattice support* always means membership in some such H ; it does not mean finite support or a discrete lattice. For instance, $\mathbb{Z} + \sqrt{2}\mathbb{Z}$ is allowed when it is a subgroup of Γ .

Theorem 1.1 (Maximal hidden negative inertia; preview). *For each integer $r \geq 1$, each nonzero divisible ordered abelian group Γ , and each $\delta \in \Gamma_{>0}$, there are explicit real Hahn series $h_1, \dots, h_r$ of positive valuation such that, with $h = (h_1, \dots, h_r)^T$ and $\varepsilon = t^\delta$,*

$$A_r = \begin{pmatrix} 1 & h^T \\ h & hh^T - \varepsilon^2 I_r \end{pmatrix} \quad (2)$$

has the following properties.

- (i) $z^* A_r z > 0$ for every nonzero $z \in \mathcal{P}_\Gamma^{r+1}$, and also for every nonzero vector over the valuation closure of $\mathcal{P}_\Gamma$ in K_Γ .
- (ii) Over K_Γ , its inertia is $(1, r, 0)$ and $\det A_r = (-\varepsilon^2)^r$.
- (iii) The vectors $(-h_j, e_j)$ have quadratic value $-\varepsilon^2$.
- (iv) The h_j can be chosen from one explicitly described family of cardinality $2^{\aleph_0}$ that is algebraically independent over $\mathcal{P}_\Gamma$.

The proof appears in Sections 5–7. The word *hidden* is restricted to vector probes from the specified subfield. It does not mean hidden from determinants, symbolic elimination, or an algorithm allowed to use the entries of A_r as new scalars. Indeed, every principal 2×2 minor containing the first coordinate is $-\varepsilon^2$. The matrix is an extremely transparent counterexample once its full entries are available.

1.2 Why primes enter

For the increasing sequence of primes $p_1, p_2, \dots$, put

$$q_p = 1 - \frac{1}{p}, \quad h = \sum_{p \in \mathcal{P}} t^{q_p \delta}, \quad P_N = \prod_{j=1}^N p_j. \quad (3)$$

A finitely generated exponent group containing δ meets $\mathbb{Q}\delta$ in $\frac{1}{m}\mathbb{Z}\delta$ for some integer $m \geq 1$. A prime not dividing m therefore supplies a coefficient of h that cannot be canceled by any series supported in that group.

More quantitatively, the least possible degree of an algebraic approximant a over $\mathbb{C}(t^\delta)$ satisfying

$$v(h - a) > q_{p_N} \delta$$

is exactly P_N ; see Theorem 4.3. Thus the obstruction is not merely that h is transcendental. There is an exact field-degree price for crossing each successive precision threshold.

1.3 A matrix-valued measure result

The scalar Hahn-measure null-ideal criterion is already present in the repository. Its Herglotz report explicitly excludes a matrix-valued null-ideal criterion from its scope [9]. We supply the *two-scale* case of that excluded criterion, not the full arbitrary-support problem. (The exact wording of the exclusion, and the precise relation to the scalar criterion, are in Section 1.6.)

Let M_0, M_1 be ordinary finite Hermitian matrix-valued measures on a measurable space. For an ordinary vector $u \in \mathbb{C}^d$, write

$$\mu_u = u^* M_0 u, \quad \nu_u = u^* M_1 u, \quad \sigma_u = M_1 u.$$

Theorem 10.1 proves that $M_0 + \varepsilon M_1$ is positive as a Hahn-valued matrix measure exactly when M_0 is positive and, for every ordinary u ,

$$\nu_u^- \ll \mu_u, \quad |\sigma_u| \ll \mu_u + \nu_u^+. \quad (4)$$

Here $\nu_u^\pm$ are the ordinary Jordan parts, $|\sigma_u|$ is ordinary vector-measure total variation, and $\ll$ means absolute continuity of null sets. The second condition is genuinely matrix-valued. It rules out coupling into a direction whose leading and first residual quadratic values both vanish.

1.4 Novelty boundary and evidence

The candidate contribution is the combined explicit package: the exact primordial approximation law, the simultaneous separation estimate for almost-disjoint prime tails over the entire finite-lattice field, maximal hidden negative inertia persisting through valuation closure and workspace enlargement, and the two-scale matrix-measure null-ideal formulation. We did not locate these combined statements in the targeted repository and literature review. That is not a certification of priority.

Hahn fields, support-defined subfields, character automorphisms, and Puiseux constructions are established subjects [3, 4, 5]. The necessary background is proved or stated explicitly below. Nor is the bare assertion that a nondense subfield can miss a negative quadratic value new mathematics: Section 8 isolates the elementary density principle so that it is not confused with the quantitative construction.

The distinction between mathematics and evidence is important here. The proofs, the exact finite checks, and any future Lean development are three different things. This manuscript supplies the first two; it does not supply a Lean verification. Appendix A records the repository snapshot and the limited scope of the searches.

1.5 Conventions: one meaning per word

This subsection and the next were written when the manuscript joined the collection (Appendix A.1); they fix vocabulary and state relations, and add no mathematics.

Positive. For a Hermitian matrix A over K_Γ (or over $L[i]$ for an ordered field L , as in Lemma 9.1 and Proposition 8.1), *positive semidefinite*, written $A \geq 0$, always means (1) with z ranging over *all* vectors over the whole field, and *positive definite*, $A > 0$, means strict inequality for every nonzero such z . This is the meaning fixed in the collection’s spectral-theory report (prop:positive of surcomplex/spectral-theory). When only a restricted set of test vectors is allowed, the statement names that set: “ $z^*Az > 0$ for every nonzero $z \in E^n$ ”, which we also call *E-probe positivity*. It is never abbreviated to “positive”: a Hermitian matrix or form called “positive” without a named probe set is positive semidefinite, and “strictly positive” without one means positive definite. The matrices A_r of Theorem 6.1 are $\mathcal{P}_\Gamma$ -probe positive and $\overline{\mathcal{P}}_\Gamma$ -probe positive, and are *not* positive semidefinite. A Hahn-valued matrix measure is *positive* when every event matrix is positive semidefinite in this sense, (32); for $d = 1$ this is the scalar meaning $\mu(E) \geq 0$ used by the Hahn–Herglotz and measures reports. An element of F_Γ is positive when its leading coefficient is. “Positive valuation” and “positive order” refer to exponents, not to signs. “Positive” never means entrywise positivity of a matrix, its meaning in the collection’s Markov-generator and matrix-scaling reports (surreal/markov-generators-at-every-scale, surreal/matrix-scaling-at-surreal-scales).

Hidden. A negative direction is *hidden* only relative to a named probe set, as in Section 1. The “hidden negative mass” of surreal/hahn-valued-measures-and-probability is a different phenomenon: a Hahn measure that passes a family of cylinder or moment tests and is negative on some event. No result transfers between the two uses of the word.

Symbols. The source wrote $\mathcal{E}_\Gamma$ for the valuation closure of $\mathcal{P}_\Gamma$; here it is $\overline{\mathcal{P}}_\Gamma$, because the Hahn–Herglotz report uses $\mathcal{E}_\Gamma(X)$ for $C(X, \mathbb{C})((t^\Gamma))$. In Sections 1–8, $\varepsilon = t^\delta$, with δ the scale of the prime tails; in Sections 9 and 10, $\varepsilon = t^\eta$ for an arbitrary $\eta > 0$. The two parts are logically independent. The letters h, h_j, h_A denote prime tails; they are not the Hankel subdiscriminants h_k of the spectral-theory report. P_N is a primorial, P (in Sections 9–10) an ordinary Hermitian matrix, $\mathcal{P}_\Gamma$ the probe field, and $\mathcal{P}$ the set of primes. Hahn exponents increase, and t^γ corresponds to $\omega^{-\gamma}$, the collection’s common convention (NOTATION.md).

1.6 Relation to other reports of the collection

The source compared itself with the collection at its pin; that comparison is kept in Appendix A. The relations below were checked at placement and revisited during the main-text proof review against repository commit b6819f7. The historical search record remains in Appendix A.

Spectral theory. Everything this report uses about one Hermitian matrix is finite Hermitian algebra over a real-closed field, as developed in surcomplex/spectral-theory: the order $\geq$ and the congruence invariance of inertia are its prop:positive, and its spec:thm:hermitian diagonalizes every Hermitian matrix over K_Γ by a unitary, for arbitrary Γ , with every exponent of the unitary and of the eigenvalues in the subgroup generated by the exponents of the entries. Theorem 6.1 is consistent with that exponent statement and shows that the subgroup in it cannot in general be replaced by a finitely generated one. The exponents of the entries of A_r generate a subgroup of Γ that contains $q_p\delta$ for infinitely many primes p ; its intersection with $\mathbb{Q}\delta$ is therefore not cyclic, so the subgroup is not finitely generated. The negative eigenvectors supplied by spec:thm:hermitian have coordinates in the Hahn field of that subgroup, and Theorem 7.1 shows that no vector with coordinates in $\overline{\mathcal{P}}_\Gamma$ can take their place. The congruence (22) uses the inertia invariance of prop:positive. Its triangular matrix T is not unitary, so this calculation does not supply the unitary eigenbasis of spec:thm:hermitian. Nothing here extends or contradicts those results. The support-index bound, Lemma 3.2, is the counterpart over intermediate fields $\mathbb{C}(t^G) \subseteq L \subseteq \mathbb{C}((t^G))$ of spec:lem:finiteindex, which computes $[K_\Lambda : K_\Gamma] = |\Lambda/\Gamma|$ for

finite Λ/Γ with the same diagonal character action; and `spec:warn:degreedrop` (relative degrees can drop in a larger workspace) is the phenomenon behind the remark that the primorial law concerns one fixed ground field (Remark 4.5).

Hahn–Herglotz positivity and Hahn-valued measures. In the subsection “Remaining scope boundaries” (Section 8.3) of `surcomplex/hahn-herglotz-positivity`, that report says: “A matrix-valued version of the scalar null-ideal criterion would have to track kernel directions of coefficient measures as well as their scalar null sets; it is not supplied here.” Its consolidated non-claims (`herg:sec:nonclaims`) repeat: “No matrix-valued null-ideal criterion is supplied.” That sentence is a non-claim, not a posed problem. Theorem 10.1 supplies the case of the two exponents $0 < \eta$, and its condition (iii) is exactly the tracking of kernel directions the sentence anticipates: $\sigma_u = M_1 u$ records where M_1 moves a direction u that is null for μ_u and for ν_u^+ . Three differences of setting must be kept in view. (a) The Herglotz measures are real-valued scalar coefficientwise Hahn measures with finite signed regular Borel coefficients on a compact Hausdorff space (`herg:def:measure`); here the coefficients are finite complex Hermitian matrix measures on an arbitrary measurable space, with no regularity. (b) The Herglotz criterion `herg:thm:nullideal` allows any well-ordered support; here there are exactly two exponents, and Question 11.1 (the transfinite matrix case) remains open. (c) Positivity here is tested with vectors in K_{Γ}^d . For $d = 1$, Corollary 10.3 is `herg:cor:twoscale`, the two-exponent case of the transfinite null-ideal criterion, which is proved once in the collection as `meas:thm:nullideal` of `surreal/hahn-valued-measures-and-probability`. `FORMALIZATION.md` records a Lean proof of the scalar two-scale criterion over any measurable space (`Surreal.NullIdeal.twoScale`, `twoScale_measure`); no matrix statement of this report is covered by it. The null-ideal results compared here use coefficientwise countable additivity; they do not require strong Hahn additivity.

Tail-spans and differential transcendence. `surreal/tail-spans-and-differential-transcendence` also builds continuum many algebraically independent Hahn tails from an explicit binary-prefix almost-disjoint family (`tail:eq:binarycode`, `tail:eq:Aalpha`), using the “private indices” of each member exactly as Theorems 5.1 and 5.2 use private primes; the coding of words differs (integers there, primes here). The mechanisms and bases are different, and neither result implies the other. There, the tails $\xi_A = \sum_{n \in A} \sqrt{p_n} t^n$ (`tail:eq:xiA`, with $t = \omega^{-1}$) have integer exponents and square-root coefficients, and are independent, even differentially for the Berarducci–Mantova derivation, over $B_0 = \mathbb{Q}((t^{\mathbb{Q}}))$ (`tail:thm:surreal`); the proof changes coefficient signs. Here the tails have coefficient one and prime-denominator exponents, and are independent over $\mathcal{P}_{\Gamma}$; the proof twists exponents by characters. For $\Gamma = \mathbb{Q}$ and $\delta = 1$ each base contains the other’s witnesses: $h_A \in \mathbb{Q}((t^{\mathbb{Q}})) = B_0$, while ξ_A has support in $\mathbb{Z}$ and so lies in $\mathcal{P}_{\mathbb{Q}}$. In particular, square-root coefficient tails cannot produce hidden directions relative to $\mathcal{P}_{\Gamma}$: with ξ_j in place of h_j , the negative vectors $(-\xi_j, e_j)$ of (21) are probe vectors. Likewise `tail:thm:closure` shows that $\sum_n \sqrt{p_n} t^n$ is a valuation limit of its algebraic truncations when $\Gamma = \mathbb{Q}$. Those truncations are algebraic over B_0 : their finitely many coefficients generate one finite extension of $\mathbb{Q}$. Merely having algebraic coefficients would not suffice, as the full transcendental tail itself demonstrates. Here Proposition 3.4 and Theorem 4.2 exclude convergence even of nets of elements algebraic over $\mathbb{C}(t^{\gamma} : \gamma \in \Gamma)$; Corollary 4.4 specializes to $\mathbb{C}(t^{\delta})$. The base fields in these algebraicity assertions differ. The warning `tail:rem:indexfamily` applies verbatim: the family (h_A) indexed by *all* infinite sets of primes is not independent, since $h_{A \cup B} + h_{A \cap B} = h_A + h_B$ is a nontrivial linear relation among four members whenever $A \cap B$ is infinite and neither set contains the other; Theorem 5.2 is about the binary-prefix family only. The upper bound in Corollary 5.3 is the same cardinality count as in `tail:thm:surreal`.

Reports placed in the same batch. `surreal/transcendence-over-bounded-support`, placed in the same commit as this report, proves algebraic independence over the fraction field of the ring

$\mathcal{B}_\Gamma(K)$ of Hahn series with support bounded above. For $\Gamma = \mathbb{Q}$ the two bases are incomparable. Every prime tail h_A has support bounded above by δ and so lies in $\mathcal{B}_\mathbb{Q}(\mathbb{C})$, but not in $\mathcal{P}_\mathbb{Q}$ by the same fresh-prime argument as Theorem 4.2. In the reverse direction, $\sqrt{1+t} \in \mathbb{C}[[t]] \subseteq \mathcal{P}_\mathbb{Q}$ does not belong to the bounded-support fraction field. Indeed, `bst:cor:onescale` gives

$$\mathbb{C}((t)) \cap \text{Frac } \mathcal{B}_\mathbb{Q}(\mathbb{C}) = \mathbb{C}(t),$$

and $1+t$ is not a square in $\mathbb{C}(t)$: the multiplicity of the factor $t+1$ in a rational square is even.

For this special intersection, the descent argument is short. If $f = a/b \in \mathbb{C}((t))$ with bounded-support a, b and $b \neq 0$, select a coset $r + \mathbb{Z}$ meeting $\text{supp } b$. Restrict $bf = a$ to that coset and shift by $-r$. Since f has integer exponents, this gives $b_r f = a_r$. The shifted supports of a_r, b_r are well-ordered, bounded subsets of $\mathbb{Z}$, hence finite, and $b_r \neq 0$. Thus f is rational in t . In particular, unbounded support alone would not suffice: the geometric series $\sum_{n \geq 1} t^n = t/(1-t)$ is already in the fraction field. No incomparability for arbitrary Γ is asserted. If Γ has no order unit (no $u > 0$ whose positive integer multiples are cofinal), each finitely generated subgroup is bounded above in Γ . Indeed, its generators are dominated in absolute value by some $u > 0$, and failure of cofinality supplies a bound above all nu . Thus $\mathcal{P}_\Gamma \subseteq \mathcal{B}_\Gamma(\mathbb{C})$. These distinctions of base and witnesses prevent either independence theorem from being identified with the other.

Background inputs. The positive-support (Neumann) lemma is `found:sub:positivesupport` of `foundations-and-computation/foundations`; algebraic closedness of K_Γ for divisible Γ , real closedness of F_Γ , and the fact that a polynomial split in a workspace gains no roots in a larger one are `polynomial:prop:workspace` of `surcomplex/polynomial-algebra`. The complex Levi–Civita field of Section 7 is the complexification of the real field defined in the same way in `foundations-and-computation/computable-surreals`. The infinite-dimensional Hahn–Hilbert spaces of `surcomplex/infinite-dimensional-hahn-spectral-theory` are not used; every matrix here is finite.

2 Hahn conventions and the probe field

2.1 Fields, valuation, conjugation, and positivity

Unless stated otherwise, Γ is a nonzero set-sized divisible ordered abelian group. We set

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)).$$

An element is a formal sum $f = \sum_{\gamma \in S} f_\gamma t^\gamma$ with well-ordered support $S \subseteq \Gamma$. For $f \neq 0$, $v(f) = \min S$; we set $v(0) = +\infty$. We use the usual conventions for comparisons involving $+\infty$. The intrinsic Γ -valuation topology has neighborhoods $\{x : v(x - a) > \rho\}$ at a , for $\rho \in \Gamma$.

The order on F_Γ is the sign of the leading real coefficient. Complex conjugation acts coefficientwise and fixes precisely F_Γ . For $z \in K_\Gamma$, write $|z|^2 = z\bar{z}$; this notation does not require choosing a square root. For $z \neq 0$,

$$|z|^2 > 0, \quad v(|z|^2) = 2v(z). \quad (5)$$

If $z_1, \dots, z_n$ are not all zero, positivity of the leading coefficients gives

$$v\left(\sum_{j=1}^n |z_j|^2\right) = 2 \min_j v(z_j). \quad (6)$$

There is no cancellation among these leading coefficients.

The standard Hahn-field closedness theorem says that K_Γ is algebraically closed and F_Γ is real closed under these hypotheses. We use this established input, rather than claim another proof of it; it belongs to the classical generalized-series theory [3], and the repository separately develops the relevant Hahn closedness machinery [7]. (In the collection it is `polynomial:prop:workspace` of `surcomplex/polynomial-algebra`, which cites Poonen for it.)

A Hermitian matrix $A \in M_n(K_\Gamma)$ is positive semidefinite if $z^*Az \geq 0$ for all $z \in K_\Gamma^n$. Its inertia $\text{In}(A) = (n_+, n_-, n_0)$ records its numbers of positive, negative, and zero squares under congruence over K_Γ . Finite-dimensional Hermitian diagonalization over a real-closed base makes this well-defined; the repository's spectral report treats this background [8] (`prop:positive` and `spec:thm:hermitian` there). This is the only meaning of ≥ 0 in this report (Section 1.5).

2.2 What summation means

A family of Hahn series is *strongly summable* when the union of its supports is well ordered and each exponent receives a contribution from only finitely many members. Products and the infinite sums used here are understood in this sense. The usual positive-support lemma licenses power-series substitution at positive-order elements [2]; in the collection it is `found:sub:positivesupport`.

Strong summation is not asserted to be convergence of finite subsums in the valuation topology. In particular, the exponents $q_p\delta$ in (3) are all below δ . Their terms form a valid Hahn sum even though its finite partial sums do not approach it to valuation greater than δ . This bounded sequence of exponents is precisely what the construction needs.

All field operations and supports are set-sized. When an actual surreal application is made, each finite collection of inputs is placed in a set-sized exponent workspace before any Hahn argument is applied.

2.3 The finite-lattice-supported subfield

Definition 2.1. The complex and real finite-lattice-supported fields are

$$\mathcal{P}_\Gamma = \bigcup_{\substack{H \leq \Gamma \\ H \text{ finitely generated}}} \mathbb{C}((t^H)), \quad \mathcal{E}_\Gamma = \bigcup_{\substack{H \leq \Gamma \\ H \text{ finitely generated}}} \mathbb{R}((t^H)). \quad (7)$$

A *probe vector* is a vector with coordinates in $\mathcal{P}_\Gamma$.

A finitely generated subgroup of an ordered abelian group is a finite-rank free abelian group. It need not be convex, discrete, or cofinal. The union in (7) is directed: the sum of finitely many finitely generated subgroups is finitely generated. As each $\mathbb{C}((t^H))$ is a subfield of K_Γ , $\mathcal{P}_\Gamma$ is a field. It contains every complex constant and every monomial t^γ . It is closed under arbitrary restrictions of a support, and

$$\mathcal{P}_\Gamma = \mathcal{E}_\Gamma[i].$$

This is an instance of the established support-family approach to Hahn subfields, not a new field construction [4].

Example 2.2. When $\Gamma = \mathbb{Q}$, every finitely generated subgroup is contained in $\frac{1}{n}\mathbb{Z}$ for a suitable integer $n \geq 1$, by taking a common denominator of its generators. Consequently

$$\mathcal{P}_\mathbb{Q} = \bigcup_{n \geq 1} \mathbb{C}((t^{1/n})),$$

the ordinary complex Puiseux-series field. Its elements can have infinite support, but each individual element has bounded denominators. For general Γ , infinitely many exponents in a fixed rank-two or rank-ten subgroup are just as admissible.

Proposition 2.3 (Closure under positive-order formal evaluation). *Let $a_1, \dots, a_m \in \mathcal{P}_\Gamma$ have positive valuation. Every formal series $\Phi \in \mathbb{C}[[X_1, \dots, X_m]]$ has a defined Hahn evaluation $\Phi(a_1, \dots, a_m)$, and this value belongs to $\mathcal{P}_\Gamma$.*

Proof. Choose one finitely generated subgroup H containing all the input supports. Each nonzero input has entirely positive support because its least exponent is positive. Their finite union is well ordered. The positive-support lemma says that the monoid of finite sums from this union is well ordered, with only finitely many words giving any prescribed sum, over all word lengths. Tagging the letters of such a word by the m input variables still gives finitely many choices. Thus all formal monomial contributions are strongly summable, regardless of the coefficients of Φ . Every resulting exponent is a finite sum of input exponents and hence lies in H . The evaluated series therefore belongs to $\mathbb{C}((t^H))$. Here a zero argument simply kills every monomial using that variable; the constant monomial contributes at exponent zero. No convergence of successive total-degree truncations is needed, even when the positive input valuations are not cofinal in Γ . $\square$

Thus infinitesimal exponential and logarithm do not leave the probe field. This assertion concerns formal evaluation at positive-order arguments; it is not a claim that every version of surreal exponentiation preserves this field.

3 Algebraic support and character orbits

This section supplies an algebraic support bound with all details needed later. The underlying diagonal character automorphisms are standard in Hahn-field automorphism theory [5]. Related finite-index monomial character arguments are also already used by the repository [8] (`spec:lem:finiteindex`, whose Galois group $\text{Hom}(\Lambda/\Gamma, \mu_q)$ acts by the same diagonal twists). They are tools here, not novelty claims.

3.1 Extending characters

Lemma 3.1 (Character extension and separation). *Let B be a subgroup of an abelian group A . Every homomorphism $B \rightarrow \mathbb{C}^\times$ extends to a homomorphism $A \rightarrow \mathbb{C}^\times$. Moreover, characters separate the elements of any abelian group.*

Proof. The group $\mathbb{C}^\times$ is divisible: every element has an n th root for each $n \geq 1$. Order the extensions of the given character by inclusion of their domains. The union along a chain is an extension, so a maximal one exists by Zorn's lemma.

If its domain B' omits $a \in A$, consider $B' + \mathbb{Z}a$. If no positive multiple of a lies in B' , assign any nonzero complex value to a . Otherwise let n be the least positive integer with $na \in B'$ and choose $c \in \mathbb{C}^\times$ with $c^n = \chi(na)$. The prescription $\chi(b + ka) = \chi(b)c^k$ is well-defined and extends the character. For if $b + ka = b' + k'a$, then $(k - k')a \in B'$. In the first case this forces $k = k'$ and $b = b'$. In the second, division with remainder by n shows $k - k' = \ell n$ for some integer ℓ ; consequently $\chi(b' - b) = \chi(na)^\ell = c^{k-k'}$, which gives the same assigned value. The formula respects addition because χ is a homomorphism on B' . Both cases contradict maximality. Hence the maximal domain is A .

For separation, define a character nontrivial on a given nonzero element on the cyclic group it generates, using a primitive root of unity when its order is finite and, for example, the value 2 when its order is infinite. Extend it to the entire group. $\square$

For a subgroup $G \leq \Gamma$ and a character $\chi : \Gamma/G \rightarrow \mathbb{C}^\times$, define

$$T_\chi \left(\sum a_\gamma t^\gamma \right) = \sum a_\gamma \chi(\gamma + G) t^\gamma. \quad (8)$$

The support is unchanged. Addition is immediate and multiplication follows from $\chi(\alpha + \beta) = \chi(\alpha)\chi(\beta)$ in each finite coefficient convolution. Thus T_χ is a field automorphism with inverse $T_{\chi^{-1}}$, and it fixes $\mathbb{C}((t^G))$ pointwise.

3.2 The support-index bound

Lemma 3.2 (Algebraic degree bounds the support index). *Let $G \leq \Gamma$ be finitely generated. Suppose $f \in K_\Gamma$ is algebraic over a field L satisfying*

$$\mathbb{C}(t^G) \subseteq L \subseteq \mathbb{C}((t^G)),$$

and let $D = [L(f) : L]$. Put $B = G + \langle \text{supp } f \rangle$. Then

$$[B : G] \leq D, \quad \text{supp } f \subseteq \frac{1}{D!}G. \quad (9)$$

The notation $\frac{1}{D!}G$ is inside the divisible group Γ . No assertion that $[B : G]$ divides D is made.

Proof. Every $T_\chi(f)$ is a root of the same minimal polynomial over L ; there are at most D distinct such images. Fix $\gamma \in \text{supp } f$. If $\gamma + G$ has infinite order, characters of its cyclic subgroup give arbitrarily many different values at that class, and Lemma 3.1 extends all of them. The coefficient at γ then gives more than D distinct images of f , a contradiction. If the class has finite order $m > D$, its m possible character values give the same contradiction. Hence its order is at most D , so $D!\gamma \in G$.

It follows that B/G is a subgroup of $(\frac{1}{D!}G)/G$, a finite group because G is finitely generated. More explicitly, if $G \cong \mathbb{Z}^k$, this quotient is isomorphic to $(\mathbb{Z}/D!\mathbb{Z})^k$: multiplication by $D!$ identifies it with $G/(D!)G$. This includes $G = 0$ and $D = 1$. Write $e = [B : G]$. A finite abelian group of order e has exactly e complex characters, as is seen by decomposing it into finite cyclic groups. Each character extends from B/G to Γ/G . Two different characters of B/G differ on at least one support exponent, because the classes of those exponents generate B/G . Their images of f are therefore different. Thus $e \leq D$. $\square$

Remark 3.3. The second bound in (9) is deliberately coarse. The first bound remembers the entire finite quotient generated by the support. In the prime construction it is the first bound, not the factorial estimate, that produces the sharp answer.

Proposition 3.4 (Algebraic closedness of the probe field). *For divisible Γ , $\mathcal{P}_\Gamma$ is algebraically closed and $\mathcal{E}_\Gamma$ is real closed. In particular every element of K_Γ algebraic over the monomial field*

$$\mathbb{C}(t^\gamma : \gamma \in \Gamma)$$

belongs to $\mathcal{P}_\Gamma$.

Proof. A nonconstant polynomial over $\mathcal{P}_\Gamma$ has finitely many coefficients. Choose a single finitely generated G containing all their supports. A root f exists in K_Γ by the standard Hahn closedness theorem. It is algebraic over $\mathbb{C}((t^G))$, so Lemma 3.2 places its support in $\frac{1}{D!}G$, again a finitely generated group. Thus $f \in \mathcal{P}_\Gamma$. Every nonconstant polynomial over $\mathcal{P}_\Gamma$ therefore has a root in it.

The real field $\mathcal{E}_\Gamma$ is ordered, and $\mathcal{E}_\Gamma[i] = \mathcal{P}_\Gamma$ is algebraically closed. The usual characterization of real-closed ordered fields gives real closedness. Alternatively, positive square roots can be chosen in F_Γ

and then in $\mathcal{E}_\Gamma$ by the support bound, while an odd-degree polynomial has a real root in F_Γ to which the same bound applies.

Finally the monomial field is contained in $\mathcal{P}_\Gamma$. A polynomial over it splits in $\mathcal{P}_\Gamma$, so every one of its roots in K_Γ is already in $\mathcal{P}_\Gamma$. $\square$

This proposition strengthens the interpretation of a probe. We are not merely testing rational expressions or bounded-degree algebraic expressions. We permit an algebraically closed field, containing all monomial scales at once and arbitrary Hahn series in every individual finitely generated exponent group.

4 A prime-denominator approximation barrier

Fix $\delta > 0$ and set $s = t^\delta$. Distinct nonnegative integer multiples of δ give distinct exponents, so no nonzero polynomial over $\mathbb{C}$ vanishes at s . Thus s is transcendental and $\mathbb{C}(t^{\mathbb{Z}\delta}) = \mathbb{C}(s)$ is a rational-function field. Let $\mathcal{P}$ be the ordinary set of positive primes, in increasing order. Define

$$h_\delta = \sum_{p \in \mathcal{P}} s^{1-1/p}, \quad h_{\delta,N} = \sum_{j=1}^N s^{1-1/p_j}. \quad (10)$$

The support of h_δ is well ordered of order type ω . Its valuation is $\delta/2$. Every support exponent is strictly less than δ .

4.1 A denominator lemma

Lemma 4.1. *If $H \leq \Gamma$ is finitely generated and contains δ , then*

$$H \cap \mathbb{Q}\delta = \frac{1}{m}\mathbb{Z}\delta \quad (11)$$

for some integer $m \geq 1$. If p is prime and $p \nmid m$, then $q_p\delta \notin H$, and for every other prime q ,

$$q_p\delta + H \neq q_q\delta + H. \quad (12)$$

Proof. The intersection is a subgroup of the finitely generated abelian group H and hence is finitely generated. As a subgroup of the rational line $\mathbb{Q}\delta$, it is cyclic. Since it contains δ , its positive generator is δ/m for an integer $m \geq 1$.

The rational number $1 - 1/p$ has denominator p in lowest terms. If $p \nmid m$, it does not belong to $\frac{1}{m}\mathbb{Z}$. For distinct primes p, q , the difference

$$q_p - q_q = \frac{p - q}{pq}$$

has a denominator divisible by p after reduction, since $p \nmid q$ and $p \nmid (p - q)$. It therefore also fails to belong to $\frac{1}{m}\mathbb{Z}$, proving (12). $\square$

Theorem 4.2 (Uniform finite-lattice approximation barrier). *For every $a \in \mathcal{P}_\Gamma$,*

$$v(h_\delta - a) < \delta. \quad (13)$$

In particular $h_\delta \notin \mathcal{P}_\Gamma$, and no element algebraic over the field of all monomials can approximate h_δ to valuation δ or higher.

Proof. Choose a finitely generated group H containing δ and $\text{supp } a$. Let m be as in Lemma 4.1 and choose a prime $p \nmid m$. The coefficient of a at $q_p \delta$ is zero, whereas that of h_δ is one. Thus $h_\delta - a$ is nonzero and

$$v(h_\delta - a) \leq q_p \delta < \delta.$$

The last assertion follows from Proposition 3.4. $\square$

4.2 The exact field-degree cost

Theorem 4.3 (Primorial approximation law). *For $N \geq 1$, let $P_N = \prod_{j=1}^N p_j$. Among all $a \in K_\Gamma$ algebraic over $\mathbb{C}(s)$ and satisfying*

$$v(h_\delta - a) > q_{P_N} \delta, \quad (14)$$

the minimum possible degree $[\mathbb{C}(s, a) : \mathbb{C}(s)]$ is exactly P_N . The finite sum $h_{\delta, N}$ attains this minimum and satisfies

$$v(h_\delta - h_{\delta, N}) = q_{P_{N+1}} \delta. \quad (15)$$

The same minimum holds with $\mathbb{R}(s)$ and real approximants.

Proof. Let a satisfy (14) and let $D = [\mathbb{C}(s, a) : \mathbb{C}(s)]$. At every exponent $q_{p_j} \delta$ for $j \leq N$, the coefficient of a must be one. Apply Lemma 3.2 with $G = \mathbb{Z}\delta$. The subgroup of Γ/G generated by these support classes contains

$$\frac{\delta}{p_1} + G, \dots, \frac{\delta}{p_N} + G,$$

since $q_p \delta \equiv -\delta/p \pmod{G}$. These classes generate $(\frac{1}{P_N} \mathbb{Z}\delta)/(\mathbb{Z}\delta)$: the integers $P_N/p_1, \dots, P_N/p_N$ have greatest common divisor one, so Bézout's identity produces δ/P_N from the δ/p_j . This quotient has order P_N , because δ has infinite order. Therefore $D \geq P_N$.

On the other hand, put $u = t^{\delta/P_N}$. Then $u^{P_N} = s$ and $h_{\delta, N} = \sum_{j=1}^N u^{P_N - P_N/p_j}$ lies in $\mathbb{C}(u)$, an extension of $\mathbb{C}(s)$ of degree at most P_N . Its support contains the first N exponents, so the same lower bound shows that its degree is exactly P_N . Its difference from h_δ has least exponent $q_{P_{N+1}} \delta$, which proves both admissibility and (15).

For real approximants, adjoining i can only decrease field degree: an $\mathbb{R}(s)$ -basis of $\mathbb{R}(s, a)$ spans $\mathbb{C}(s, a)$ over $\mathbb{C}(s)$. Thus $[\mathbb{C}(s, a) : \mathbb{C}(s)] \leq [\mathbb{R}(s, a) : \mathbb{R}(s)]$, so the same lower bound applies. The displayed real finite sum has real degree at most P_N and hence exactly P_N . $\square$

N	p_N	Precision crossed, divided by δ	Least degree P_N
1	2	1/2	2
2	3	2/3	6
3	5	4/5	30
4	7	6/7	210
5	11	10/11	2310
6	13	12/13	30030

The inequalities in the theorem are strict. Crossing the N th support exponent means matching its coefficient as well. Merely reaching that valuation without crossing it need not match the N th coefficient.

Corollary 4.4 (A degree-limited obstruction). *If a is algebraic over $\mathbb{C}(s)$ of degree less than P_N , then*

$$v(h_\delta - a) \leq q_{P_N} \delta.$$

No sequence of algebraic approximants, regardless of the growth of its degrees, converges to h_δ in the intrinsic Γ -valuation topology.

Proof. The first assertion is the contrapositive of Theorem 4.3. The second follows from Theorem 4.2: every such approximant stays outside the valuation neighborhood $v(h_\delta - a) > \delta$. $\square$

The same fixed neighborhood excludes convergence of any net of these approximants. Increasing degrees or replacing sequences by nets cannot cross the missing valuation scale.

Remark 4.5 (What the degree law does not measure). A field degree is not a lower bound for expression length or running time. The sum $h_{\delta,N}$ has a short expression involving N radicals, despite its primordial degree over $\mathbb{C}(s)$. If the ground field already contains those radicals, its degree over the new ground field is one. The sharpness claim concerns exactly the field and precision specified in Theorem 4.3. The same drop of relative degree under enlargement of the base is recorded for Hahn fields in `spec:warn:degreedrop` of the spectral-theory report.

5 Almost-disjoint prime tails

The single tail suffices for a 2×2 counterexample. To hide many independent negative directions at once, we need a simultaneous separation estimate that allows arbitrary coefficients from $\mathcal{P}_\Gamma$. Algebraic independence alone would not give the required valuation bound.

5.1 An explicit continuum-sized family

For an infinite set $A \subseteq \mathcal{P}$, define

$$h_A = \sum_{p \in A} t^{q_p \delta}. \quad (16)$$

Every such sum is a positive-order real Hahn series. Choose a bijection from the set of nonempty finite binary words to the primes, listing words first by length and then lexicographically. For an infinite binary sequence $b \in \{0, 1\}^\mathbb{N}$, let A_b consist of the primes assigned to its nonempty finite prefixes. Each A_b is infinite. Distinct branches have only finitely many common prefixes, so $A_b \cap A_c$ is finite when $b \neq c$.

We shall only use the following property of this family: among finitely many distinct selected sets, each one has infinitely many primes that belong to none of the other selected sets. The construction makes the family explicit without choosing a transcendence basis. The same binary-prefix device, with words coded by integers instead of primes, is `tail:eq:Aalpha` of the tail-spans report (Section 1.6).

5.2 The fresh-prime separation estimate

Theorem 5.1 (Uniform linear-combination separation). *Let $A_1, \dots, A_r$ be infinite pairwise almost-disjoint sets of primes, and write $h_j = h_{A_j}$. If $a_0, a_1, \dots, a_r \in \mathcal{P}_\Gamma$ and $(a_1, \dots, a_r) \neq 0$, then*

$$v\left(a_0 + \sum_{j=1}^r h_j a_j\right) < \delta + \min_{1 \leq j \leq r} v(a_j). \quad (17)$$

In particular the linear combination on the left is nonzero.

Proof. Let $m_0 = \min_{1 \leq j \leq r} v(a_j)$ and choose j with $v(a_j) = m_0$. Choose one finitely generated subgroup H containing δ and every support $\text{supp } a_i$, including $i = 0$. Write $H \cap \mathbb{Q}\delta = (\delta/m)\mathbb{Z}$.

There is a prime $p \in A_j$ that does not divide m and belongs to no other selected A_i . Consider the coefficient at

$$\alpha = m_0 + q_p \delta.$$

The term $t^{q_p \delta} a_j$ contributes the nonzero leading coefficient of a_j . No term of a_0 contributes, because $\alpha \notin H$ by Lemma 4.1. No term $t^{q_q \delta} a_i$ for a different prime q contributes, because the cosets $q_p \delta + H$ and $q_q \delta + H$ are distinct. The same prime p occurs in none of the other h_i . Within $t^{q_p \delta} a_j$, only the coefficient at m_0 can contribute to α .

Thus the coefficient at α in the entire linear combination is exactly the nonzero leading coefficient of a_j . The Hahn products are well-defined, so coefficient extraction is legitimate. Consequently the valuation of the combination is at most $\alpha = m_0 + q_p \delta < m_0 + \delta$. $\square$

The argument detects one surviving coefficient rather than estimating an infinite sum. It does not require a norm, an Archimedean valuation rank, a cofinal sequence of exponents, or convergence of partial sums. The exponent m_0 can be arbitrarily large, small, or in a different Archimedean class from δ .

5.3 Algebraic independence over the entire probe field

Theorem 5.2 (Independent prime-tail family). *For the almost-disjoint family constructed above,*

$$\{h_{A_b} : b \in \{0, 1\}^{\mathbb{N}}\}$$

is algebraically independent over $\mathcal{P}_\Gamma$.

Proof. We prove algebraic independence of every finite subfamily by induction on its size. The same argument covers the initial size one. Suppose $h_1, \dots, h_{r-1}$ are algebraically independent over $\mathcal{P}_\Gamma$, and assume that a nonzero polynomial relation involving also h_r exists. After treating the last variable separately, it becomes

$$R(h_r) = 0, \quad R(X) \in \mathcal{P}_\Gamma(h_1, \dots, h_{r-1})[X], \quad R \neq 0. \quad (18)$$

Indeed, at least one coefficient of the original polynomial in the last variable remains nonzero by the induction hypothesis. We may take R with coefficients polynomial, rather than rational, in the previous h_j . Let $d = \deg R \geq 1$.

Choose a finitely generated group H containing δ and all supports of the finitely many original coefficients from $\mathcal{P}_\Gamma$. As before, write $H \cap \mathbb{Q}\delta = (\delta/m)\mathbb{Z}$. Choose a prime

$$p \in A_r \setminus \bigcup_{j < r} A_j, \quad p \nmid m, \quad p > d.$$

Such a prime exists because only finitely many primes are excluded by the intersections, the divisors of m , and the bound d .

Let J be the subgroup of Γ generated by H and all δ/q with q in the selected prime sets and $q \neq p$. Every rational coefficient of δ in $J \cap \mathbb{Q}\delta$ has a reduced denominator not divisible by p . To see this, any expression in J is $h + \ell$, with $h \in H$ and ℓ a rational multiple of δ whose denominator uses only primes different from p . If the expression lies on the rational line, then so does h , and its denominator divides m . Thus $\delta/p \notin J$, whereas $p(\delta/p) = \delta \in J$. Its class in Γ/J has order exactly p .

Choose a primitive p th root of unity ζ and extend the character that sends $\delta/p + J$ to ζ to all of Γ/J . The associated automorphism T fixes all original coefficients, all h_j for $j < r$, and every term of h_r except its term at $q_p \delta = \delta - \delta/p$. For $k = 0, \dots, p-1$,

$$T^k(h_r) = h_r + (\zeta^{-k} - 1)t^{q_p \delta}. \quad (19)$$

These are p distinct elements, since their coefficients at $q_p \delta$ are the distinct numbers ζ^{-k} . The extension character need not have order p on all of Γ/J ; only these p values and the displayed action on h_r are

used. Since T fixes the original coefficients and every previous h_j , it fixes every coefficient of R . All the displayed elements are therefore roots of that same nonzero polynomial. This contradicts $p > d$. The induction is complete. $\square$

The theorem is stronger than algebraic independence over $\mathbb{C}$ or over a single rational-function field. Its base contains all monomials and all series supported in every individual finitely generated subgroup. The proof uses fresh prime denominators in the *exponents*, rather than algebraically independent choices of ordinary coefficients. This distinguishes it from the repository's existing tail-span and differential-transcendence constructions [7] (`tail:thm:surreal`; the comparison of bases is in Section 1.6). As `tail:rem:indexfamily` warns there, the restriction to an almost-disjoint family is needed: if A, B are infinite prime sets with infinite intersection, neither containing the other, then $h_{A \cup B} + h_{A \cap B} = h_A + h_B$ is a nontrivial linear relation.

Corollary 5.3. *For $\Gamma = \mathbb{Q}$,*

$$\text{trdeg}(\mathbb{C}((t^{\mathbb{Q}}))/\mathcal{P}_{\mathbb{Q}}) = 2^{\aleph_0}.$$

Proof. Theorem 5.2 gives the lower bound. A Hahn series on the countable group $\mathbb{Q}$ is in particular a function $\mathbb{Q} \rightarrow \mathbb{C}$. There are only $|\mathbb{C}|^{\aleph_0} = 2^{\aleph_0}$ such functions, giving the upper bound. $\square$

This cardinal conclusion is included to record what the explicit construction implies. No novelty is claimed for a generic transcendence-degree cardinality phenomenon.

6 Maximal hidden negative inertia

We now combine the uniform separation estimate with one finite congruence calculation. The separation theorem carries the substantive support information; the matrix algebra is elementary.

Theorem 6.1 (Hidden-inertia construction). *Let $A_1, \dots, A_r$ be infinite pairwise almost-disjoint prime sets, $r \geq 1$, and put $h_j = h_{A_j}$, $h = (h_1, \dots, h_r)^T$, $\varepsilon = t^\delta$. The real symmetric matrix*

$$A_r = \begin{pmatrix} 1 & h^T \\ h & hh^T - \varepsilon^2 I_r \end{pmatrix}$$

is strictly positive on every nonzero vector in $\mathcal{P}_{\Gamma}^{r+1}$. Over K_{Γ} , however,

$$\text{In}(A_r) = (1, r, 0), \quad \det A_r = (-\varepsilon^2)^r. \quad (20)$$

Its standard-part matrix is $\text{diag}(1, 0, \dots, 0)$.

Proof. For $z = (a_0, y)$ with $y = (a_1, \dots, a_r)^T$,

$$z^* A_r z = \left| a_0 + \sum_{j=1}^r h_j a_j \right|^2 - \varepsilon^2 \sum_{j=1}^r |a_j|^2. \quad (21)$$

If $y = 0$ and $z \neq 0$, this equals $|a_0|^2 > 0$. Otherwise set $m_0 = \min_j v(a_j)$. Theorem 5.1 and (5) show that the first term in (21) is positive and has valuation strictly less than $2m_0 + 2\delta$. By (6), the second term has exactly that valuation. Hence the leading term of the difference is the positive leading term of the first term. This proves strict positivity for probe vectors.

For the actual inertia, define

$$T = \begin{pmatrix} 1 & h^T \\ 0 & I_r \end{pmatrix}, \quad D = \text{diag}(1, -\varepsilon^2 I_r).$$

A direct multiplication gives

$$A_r = T^*DT, \quad \det T = 1. \quad (22)$$

Thus the inertia and determinant are those in (20). Equivalently, the r -dimensional subspace

$$\{(-h^T y, y) : y \in K_\Gamma^r\}$$

is negative definite, since its quadratic value is $-\varepsilon^2 \sum |y_j|^2$. The vector e_0 has value one. Finally every h_j and ε has positive valuation, giving the stated standard part. $\square$

Corollary 6.2 (Smallest example). *With the single all-primes tail $h = h_\delta$,*

$$A = \begin{pmatrix} 1 & h \\ h & h^2 - t^{2\delta} \end{pmatrix} \quad (23)$$

is strictly positive on $\mathcal{P}_\Gamma^2 \setminus \{0\}$ but has determinant $-t^{2\delta}$ and value $-t^{2\delta}$ at $(-h, 1)$.

Proof. For $b \neq 0$, the quadratic value at (a, b) is

$$|b|^2 \left(|a/b + h|^2 - t^{2\delta} \right).$$

Theorem 4.2 applies to $-a/b \in \mathcal{P}_\Gamma$. When $b = 0$ and $(a, b) \neq 0$, the value is $|a|^2 > 0$. The remaining statements follow directly from the determinant and quadratic identity. $\square$

6.1 Two sharpness statements

The dimension and inertia counts are optimal in the following simple, precise senses. In dimension one, testing the vector 1 detects a negative scalar, so no example exists. Thus size two is minimal. In dimension $r + 1$, a form that is positive at e_0 cannot have $r + 1$ negative directions. Our construction has r , the maximum possible number compatible with even this one positive probe.

These are sharpness claims about dimension and inertia, not about the ordinal type of the matrix's entire support. Although each h_j has support of order type ω , products $h_i h_j$ need not have the same support order type. No support-order threshold for arbitrary matrix entries is asserted.

6.2 Why algebraic closedness does not rescue the tests

The probe field is algebraically closed, but the matrix entries need not lie in it. Already $h \notin \mathcal{P}_\Gamma$ by Theorem 4.2. A transfer principle for formulas with parameters from a real-closed subfield does not allow parameters outside that subfield. There is therefore no conflict with real-closed field theory.

If the probe field is enlarged to contain the h_j , the explicit negative vectors immediately become available. Similarly, computing the principal minor on coordinates $0, j$ yields

$$\det \begin{pmatrix} 1 & h_j \\ h_j & h_j^2 - \varepsilon^2 \end{pmatrix} = -\varepsilon^2.$$

Hence the construction is not an obstruction to an exact positivity algorithm operating in the field generated by the entries. It is an obstruction to replacing quantification over that field by a particular fixed support-restricted collection of test vectors.

7 Valuation closure and the Levi–Civita case

Let $\overline{\mathcal{P}}_\Gamma$ be the closure of $\mathcal{P}_\Gamma$ inside K_Γ for the valuation topology. Explicitly, $a \in \overline{\mathcal{P}}_\Gamma$ means that for every $\rho \in \Gamma$ there exists $b \in \mathcal{P}_\Gamma$ with $v(a - b) > \rho$. This definition uses neighborhoods, not a presumed countable cofinal sequence. The closure is a subfield, since addition, multiplication, and inversion away from zero are continuous in a valued field.

Theorem 7.1 (The barrier persists through valuation closure). *For the tails in Theorem 5.1, the estimate*

$$v\left(a_0 + \sum_{j=1}^r h_j a_j\right) < \delta + \min_{1 \leq j \leq r} v(a_j)$$

continues to hold for $a_0, \dots, a_r \in \overline{\mathcal{P}}_\Gamma$, provided $(a_1, \dots, a_r) \neq 0$. Consequently A_r is strictly positive on every nonzero vector over $\overline{\mathcal{P}}_\Gamma$.

Proof. Suppose the estimate fails and write $m_0 = \min_{j \geq 1} v(a_j)$. For every $j = 0, \dots, r$, choose $b_j \in \mathcal{P}_\Gamma$ with

$$v(a_j - b_j) > m_0 + \delta.$$

For $j \geq 1$, these approximations preserve the minimum valuation: $\min_j v(b_j) = m_0$. Indeed, all errors have valuation greater than m_0 , so no b_j has valuation below m_0 , and an a_j with valuation m_0 retains its nonzero leading coefficient. In particular, $(b_1, \dots, b_r) \neq 0$. Write $L_a = a_0 + \sum h_j a_j$ and $L_b = b_0 + \sum h_j b_j$. Since each $v(h_j) > 0$,

$$v(L_a - L_b) > m_0 + \delta.$$

The assumed $v(L_a) \geq m_0 + \delta$ therefore implies $v(L_b) \geq m_0 + \delta$, contrary to Theorem 5.1. The positivity conclusion follows from exactly the valuation comparison in Theorem 6.1. $\square$

7.1 Identifying the closure for rational exponents

Definition 7.2. A subset $S \subseteq \mathbb{Q}$ is *left finite* if for every $\rho \in \mathbb{Q}$ the set $S \cap (-\infty, \rho]$ is finite. The complex Levi–Civita field is the field of complex Hahn series with left-finite rational support.

The Puiseux and Levi–Civita relationship belongs to the classical valuation-theoretic setting; the Puiseux-field conventions are discussed in [6]. We include the elementary closure argument to make the exact topology explicit.

Proposition 7.3. *The closure $\overline{\mathcal{P}}_\mathbb{Q}$ of $\mathcal{P}_\mathbb{Q}$ in $\mathbb{C}((t^\mathbb{Q}))$ is precisely the complex Levi–Civita field.*

Proof. A well-ordered subset of $\frac{1}{n}\mathbb{Z}$ is bounded below and has only finitely many elements below each rational bound. Hence every Puiseux series has left-finite support.

Suppose f lies in the closure. For each $\rho \in \mathbb{Q}$, take a Puiseux series g with $v(f - g) > \rho$. The two series have identical coefficients at all exponents at most ρ . The support of f below that bound is therefore finite. Thus f is left finite.

Conversely, for a left-finite series f and a bound ρ , truncate to the finitely many exponents at most ρ . This finite sum is a Puiseux series, and the difference has valuation greater than ρ , or is zero. Therefore f lies in the closure. $\square$

Corollary 7.4. *There are explicit Hermitian matrices over $\mathbb{C}((t^\mathbb{Q}))$ with inertia $(1, r, 0)$ that are strictly positive on every nonzero vector with complex Levi–Civita coordinates.*

Proof. Choose the almost-disjoint prime tails with $\Gamma = \mathbb{Q}$ and $\delta = 1$. Theorem 6.1 gives the inertia, and Theorem 7.1 gives strict positivity on the closure identified in Proposition 7.3. $\square$

The coefficients of these matrices are *not* all in the Levi–Civita field. In particular, h_δ has infinitely many exponents below δ when $\delta \in \mathbb{Q}_{>0}$. The corollary does not say that a Hermitian matrix defined over the Levi–Civita field can be strictly positive on all nonzero Levi–Civita vectors and become indefinite under an ordered extension. It says that Levi–Civita probes are insufficient for certain matrices defined in a larger Hahn field.

8 Density and actual surreal numbers

8.1 The elementary density principle

The following proposition explains the general mechanism independently of generalized series. It is included as an elementary organizing fact, not as a new theorem of ordered-field theory.

Proposition 8.1 (Quadratic testing characterizes order density). *Let $E \subseteq F$ be ordered fields. The following conditions are equivalent.*

- (i) *E is order dense in F .*
- (ii) *For every finite symmetric matrix over F , nonnegativity on all vectors in E^n implies nonnegativity on all vectors in F^n .*
- (iii) *The implication in (ii) holds for matrices of size two.*

If E is not order dense, a size-two form can in fact be strictly positive on every nonzero E -vector and negative at an F -vector.

Proof. Assume density and let $q(x) = x^T A x$ satisfy $q(u) < 0$ for some $u \in F^n$. We can see the needed polynomial continuity directly over any ordered field. Put

$$C = 1 + \sum_{i,j} |A_{ij}|(|u_i| + |u_j| + 1), \quad \theta = \min\left(1, \frac{-q(u)}{2C}\right) > 0.$$

If $|x_i - u_i| < \theta$ for every i , expanding $q(x) - q(u)$ gives $|q(x) - q(u)| < C\theta \leq -q(u)/2$, so $q(x) < 0$. Density supplies such an $x \in E^n$ by choosing each coordinate in $(u_i - \theta, u_i + \theta)$. This proves (i) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (iii) is immediate.

Conversely, if E is not dense, choose $a < b$ in F such that $(a, b) \cap E = \emptyset$. Put

$$h = \frac{a+b}{2}, \quad e = \frac{b-a}{4} > 0.$$

The closed interval $[h - e, h + e]$ lies inside (a, b) . For

$$q(x, y) = (x - hy)^2 - e^2 y^2,$$

if $(x, y) \in E^2$ is nonzero and $y = 0$, then $q = x^2 > 0$. If $y \neq 0$, the ratio $x/y \in E$ is outside that closed interval, so $|x/y - h| > e$ and again $q > 0$. Yet $q(h, 1) = -e^2 < 0$. This disproves (iii). $\square$

For the corresponding complexified fields $E[i] \subseteq F[i]$, the same conclusion follows by applying continuity to real and imaginary coordinates. The real witness above also works directly: if $z/w = x + iy$ with $x, y \in E$, then $|z/w - h|^2 = (x - h)^2 + y^2 > e^2$. This handles $w \neq 0$; for $w = 0$ and $z \neq 0$, the quadratic value is $|z|^2 > 0$.

This proposition is why a bare example over a nondense subfield would be too weak a contribution by itself. The prime-tail package specifies a particularly large natural probe field, gives a precise missed valuation scale, defeats its closure, constructs the maximal number of negative directions, and computes exact algebraic approximation costs. General results about dense subfields of henselian fields address a broader but different question [6]; nothing here says that a Hahn field has no proper dense subfields.

8.2 Transport to Conway normal forms

If Γ is a set-sized ordered additive subgroup of $\mathbf{No}$, the normal-form correspondence

$$\sum_{\gamma \in S} c_\gamma t^\gamma \mapsto \sum_{\gamma \in S} c_\gamma \omega^{-\gamma} \quad (24)$$

embeds its real Hahn field into $\mathbf{No}$ and its complex Hahn field into $\mathbf{No}[i]$. The minus sign is essential: increasing Hahn exponents become decreasing Conway growth exponents. The correspondence preserves addition, multiplication, conjugation, and the leading-coefficient order. These are the standard normal-form operations [1]; the repository also emphasizes this orientation issue [7] (NOTATION.md).

Take $\Gamma = \mathbb{Q}$ and $\delta = 1$. Then

$$h = \sum_{p \in \mathcal{P}} \omega^{-1+1/p} = \omega^{-1/2} + \omega^{-2/3} + \omega^{-4/5} + \omega^{-6/7} + \dots \quad (25)$$

is an actual surreal number. No limiting interpretation is required; this is its Conway normal form. The actual surcomplex matrix

$$\begin{pmatrix} 1 & h \\ h & h^2 - \omega^{-2} \end{pmatrix} \quad (26)$$

has a negative quadratic value at $(-h, 1)$ and is strictly positive on all nonzero finite-lattice-supported surcomplex probe vectors.

Theorem 8.2 (Workspace-independent surreal probe obstruction). *The positivity conclusion for (26), and the higher-dimensional construction transported by (24), holds for every finite surcomplex vector whose coordinates each have Conway support contained in some finitely generated additive subgroup of $\mathbf{No}$. The subgroup may depend on the vector and may contain arbitrarily large surreal exponents.*

Proof. For the given finite vector, choose a finitely generated subgroup H containing all its exponent supports and 1. Its divisible hull is a set-sized ordered abelian group $\Gamma' \subseteq \mathbf{No}$. Both the vector and the prime-tail matrix are represented in the Hahn workspace for Γ' , with Hahn exponents opposite to the Conway exponents. The probe coordinates lie in $\mathcal{P}_{\Gamma'}$. Theorem 6.1 applies there and transports back. $\square$

Thus adding any finitely many new monomial scales to a particular probe representation does not remove the obstruction. The quantification is local to each finite input. No set containing all surreal numbers, and no class-indexed Hahn summation, is being used.

An analogous statement applies to actual surcomplex coordinates algebraic over rational expressions in surreal monomials. To justify this without assuming a workspace already contains the root, let x be such a coordinate. Its equation uses finitely many monomials; let H be the finitely generated group of their Hahn exponents. Adjoin 1 and the negatives of all Conway exponents of x to H and take the divisible hull. This is a nonzero set-sized ordered subgroup Γ' of $\mathbf{No}$, and its Hahn workspace contains both x and the equation. Now apply Lemma 3.2 inside that workspace with $G = H$ and $L = \mathbb{C}(t^H)$. If $D = [L(x) : L]$, the support of x lies in $\frac{1}{D!}H$, which is finitely generated. Thus x is an allowed probe. For a finite vector take the sum of these finitely many support groups. This does not include a coordinate whose defining parameters already contain the prime tail itself.

9 The exact two-scale matrix criterion

We now turn to the second part of the paper. Here Γ may be any nonzero ordered abelian group; divisibility is unnecessary. Let $\varepsilon = t^\eta$ for some $\eta > 0$. The coefficients of the two matrices below are ordinary complex numbers, while vectors in the positivity test are allowed to have arbitrary Hahn coordinates.

9.1 An isotropic vector of a positive form lies in its kernel

Lemma 9.1. *Let L be an ordered field, let $A = A^*$ over $L[i]$, and assume $x^*Ax \geq 0$ for all x . If $u^*Au = 0$, then $Au = 0$.*

Proof. Fix a vector w and write $b = w^*Au$ and $c = w^*Aw \geq 0$. If $b \neq 0$, then $|b|^2 > 0$ because it is the sum of the squares of its real and imaginary parts. Choose $s = -\lambda b$ with $\lambda > 0$ in L . Hermitian symmetry gives $u^*Aw = \bar{b}$, so

$$(u + sw)^*A(u + sw) = s\bar{b} + \bar{s}b + |s|^2c = -2\lambda|b|^2 + \lambda^2|b|^2c.$$

For $c = 0$ this is negative. For $c > 0$, taking $\lambda = 1/c$ also makes it negative. Both contradict positivity. Hence $w^*Au = 0$ for every w , and the coordinate vectors give $Au = 0$. $\square$

Theorem 9.2 (Two-scale matrix positivity). *Let $P, Q \in M_d(\mathbb{C})$ be Hermitian. Write $N = \ker P$ and let C_N denote the compression of Q to N , viewed as an ordinary Hermitian operator on N . Then*

$$P + \varepsilon Q \geq 0 \iff \begin{cases} P \geq 0, \\ C_N \geq 0, \\ Qu = 0 \text{ for every } u \in \ker C_N \subseteq N. \end{cases} \quad (27)$$

Equivalently, the last two clauses say that the quadratic form of Q is nonnegative on N , and that

$$u \in N, \quad u^*Qu = 0 \implies Qu = 0. \quad (28)$$

When these conditions hold,

$$\ker(P + \varepsilon Q) = (\ker P \cap \ker Q) \otimes_{\mathbb{C}} K_\Gamma, \quad (29)$$

$$\text{rank}(P + \varepsilon Q) = \text{rank } P + \text{rank } C_N. \quad (30)$$

Proof. Suppose $P + \varepsilon Q \geq 0$. Testing an ordinary vector whose P -value is negative would give a negative leading coefficient, so $P \geq 0$. For ordinary $u \in N$, the value is εu^*Qu , proving $C_N \geq 0$. If $u \in \ker C_N$, then $u^*Qu = 0$. Its value under $P + \varepsilon Q$ is zero, so Lemma 9.1 gives $(P + \varepsilon Q)u = 0$. As $Pu = 0$, this means $Qu = 0$.

Conversely assume the three clauses. Use an ordinary unitary change of basis to decompose

$$\mathbb{C}^d = R \oplus N_+ \oplus N_0, \quad R = \text{ran } P, \quad N_+ = \text{ran } C_N, \quad N_0 = \ker C_N.$$

The matrices P_R and $C_+ = C_N \upharpoonright_{N_+}$ are positive definite on their respective spaces. By hypothesis Q annihilates N_0 , so the entire matrix is zero on that last direct summand. On $R \oplus N_+$ it has the block form

$$\begin{pmatrix} P_R + \varepsilon A & \varepsilon B \\ \varepsilon B^* & \varepsilon C_+ \end{pmatrix}. \quad (31)$$

Assume first that both active spaces have positive dimension. Let $U = P_R + \varepsilon A$. Every leading principal minor of U has positive constant coefficient, the corresponding minor of P_R . For a Hermitian matrix

over an ordered field complexification, positive leading principal minors imply positive definiteness: elimination gives an LDL^* factorization with diagonal pivots $\Delta_j/\Delta_{j-1} > 0$, where $\Delta_0 = 1$ and Δ_j is the j th leading principal minor. Its quadratic value is a sum of positive pivots times squared moduli after an invertible triangular change of coordinates. This works for all Hahn vectors and needs no real-closedness. Thus $U > 0$.

The adjugate formula shows that U^{-1} has entries rational in ε . Since $\det U$ has nonzero constant term, its reciprocal has valuation zero and reciprocal constant coefficient; hence the constant matrix of U^{-1} is P_R^{-1} . Put

$$V = \begin{pmatrix} I & -\varepsilon U^{-1}B \\ 0 & I \end{pmatrix}.$$

A block multiplication gives the explicit congruence

$$V^* \begin{pmatrix} U & \varepsilon B \\ \varepsilon B^* & \varepsilon C_+ \end{pmatrix} V = \text{diag}(U, \varepsilon(C_+ - \varepsilon B^* U^{-1} B)).$$

The second factor in parentheses is Hermitian and has constant matrix C_+ . Every leading principal minor therefore has positive constant coefficient, so the same pivot argument makes it positive definite. Since $\varepsilon > 0$ and V is invertible, the entire active block is positive definite. If $R = 0$, it is simply εC_+ ; if $N_+ = 0$, it is $P_R + \varepsilon A$. If both are zero there is no active block. These cases need no empty inverse.

The only kernel is therefore $N_0 \otimes_{\mathbb{C}} K_{\Gamma}$. Within N , the condition $Qu = 0$ implies $C_N u = 0$ by compression; the converse is exactly the assumed annihilation clause. Thus $N_0 = \ker P \cap \ker Q$, and the active dimension is $\dim R + \dim N_+ = \text{rank } P + \text{rank } C_N$. This proves (29) and (30). All inverses used are rational expressions in ε ; no sequential convergence assertion is involved. $\square$

The theorem is a finite Schur-complement criterion. Its role is to expose the exact null-direction coupling needed by the measure theorem. We do not claim Schur complements or singular perturbation theory as new techniques.

9.2 The two ways a residual direction behaves

Example 9.3 (A coupling that positivity forbids). Take

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Here $N = \mathbb{C}e_2$ and $C_N = 0$, but $Qe_2 = e_1 \neq 0$. Thus

$$P + \varepsilon Q = \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}$$

is not positive: its determinant is $-\varepsilon^2$, and $(-\varepsilon, 1)$ has value $-\varepsilon^2$. Nevertheless every ordinary vector $u = (a, b)$ has nonnegative value: if $a \neq 0$, its leading value is $|a|^2 > 0$, and if $a = 0$, its value is zero. Ordinary scalar vector tests miss the coupling completely.

Example 9.4 (A coupling that positivity allows). With the same P but

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

the compression C_N is strictly positive. Then

$$\begin{pmatrix} 1 & \varepsilon \\ \varepsilon & \varepsilon \end{pmatrix} > 0, \quad \det = \varepsilon - \varepsilon^2 > 0.$$

The ordinary matrix Q itself is indefinite. Requiring $Q \geq 0$ would therefore be too strong. Only its action on the leading kernel and the coupling out of its residual zero space are relevant.

Example 9.5 (Why more than two scales changes the question). For ordinary real c ,

$$B_c = \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & c\varepsilon^2 \end{pmatrix}$$

is congruent to $\text{diag}(1, (c-1)\varepsilon^2)$ and hence is positive semidefinite exactly when $c \geq 1$. For every $c \geq 0$, however, all ordinary vector tests are nonnegative: a vector with first coordinate nonzero has positive leading value, and one with first coordinate zero has value $c\varepsilon^2|b|^2$.

The third-scale coefficient can repair, exactly cancel, or fail to repair the second-order Schur correction. Therefore Theorem 9.2 is not an arbitrary-support matrix criterion in disguise. Its annihilation condition is forced because no third-scale coefficient is available to repair the coupling.

10 A two-scale matrix-valued null-ideal theorem

10.1 Definitions and the quantifiers

Let (X, Σ) be a measurable space. A finite Hermitian $\mathbb{C}^{d \times d}$ -valued measure is a matrix of finite complex measures $M = (M_{ij})$ with $M(E)^* = M(E)$ for every measurable E . Countable additivity is entrywise, and each entry has finite total variation. No topology on X or regularity assumption is needed.

For two such measures, define

$$M = M_0 + \varepsilon M_1.$$

This is a coefficientwise Hahn-valued matrix measure with two possible exponents. We call it *positive* when

$$z^* M(E) z \geq 0 \quad \text{for every } E \in \Sigma \text{ and every } z \in K_{\Gamma}^d. \quad (32)$$

The vector z may depend on E and may contain infinitesimals. The model is coefficientwise countable additivity. Strong Hahn summability of all countable families of event masses is not assumed. The repository distinguishes these notions, and we retain that separation [9, 7]; the distinction is drawn in *surreal/hahn-valued-measures-and-probability*.

If M_0 is an ordinary positive matrix measure, then for $u \in \mathbb{C}^d$

$$\mu_u(E) = u^* M_0(E) u \quad (33)$$

is a finite positive real measure. For M_1 , set

$$\nu_u(E) = u^* M_1(E) u, \quad \sigma_u(E) = M_1(E) u. \quad (34)$$

The first is a finite signed real measure and the second a finite complex vector measure. Write $\nu_u = \nu_u^+ - \nu_u^-$ for its Jordan decomposition, and $|\sigma_u|$ for total variation using the ordinary Euclidean norm. Any other ordinary norm on $\mathbb{C}^d$ gives the same null sets. Finiteness follows directly from the entry measures: for each event E ,

$$|\sigma_u|(E) \leq \sum_{i,j} |u_j| |(M_1)_{ij}|(E) < \infty.$$

Indeed, apply the triangle inequality to $\|M_1(B)u\|$ on each piece of a finite measurable partition of E , then take the supremum over partitions.

For a positive scalar measure λ , the assertion $|\sigma_u| \ll \lambda$ means that σ_u vanishes on every measurable subset of every λ -null set. Indeed, this is equivalent to the usual variation formulation by the definition of total variation through finite partitions. If $\sigma_u(B) = 0$ for every measurable $B \subseteq E$, each partition sum defining $|\sigma_u|(E)$ is zero. Conversely, $\|\sigma_u(B)\| \leq |\sigma_u|(B) \leq |\sigma_u|(E)$, so zero variation on E forces all those vector masses to vanish.

10.2 The characterization

Theorem 10.1 (Two-scale matrix null ideals). *Let M_0, M_1 be finite Hermitian matrix-valued measures and $\varepsilon = t^\eta$ with $\eta > 0$. Then $M_0 + \varepsilon M_1$ is positive in the sense of (32) if and only if all of the following hold:*

(i) M_0 is an ordinary positive matrix-valued measure.

(ii) For every ordinary $u \in \mathbb{C}^d$,

$$\nu_u^- \ll \mu_u. \quad (35)$$

(iii) For every ordinary $u \in \mathbb{C}^d$,

$$|\sigma_u| \ll \mu_u + \nu_u^+. \quad (36)$$

The result holds for every nonzero ordered value group Γ ; divisibility is not required. No bounded Radon–Nikodym derivative is required in either absolute-continuity condition.

Proof. Necessity. Assume M is positive. For every ordinary u and event E , a negative value $u^* M_0(E) u$ would be the negative leading coefficient of $u^* M(E) u$. Thus M_0 is positive.

Fix u and let E be μ_u -null. Every measurable $B \subseteq E$ is also μ_u -null. Positivity gives

$$0 \leq u^* M(B) u = \varepsilon \nu_u(B),$$

so ν_u restricts to a positive measure on E . For example, intersect E with a negative set in a Hahn decomposition of ν_u : its mass is $-\nu_u^-(E)$ and must be nonnegative. Thus $\nu_u^-(E) = 0$, proving (35).

Next suppose $(\mu_u + \nu_u^+)(E) = 0$. By (35), $\nu_u^-(E) = 0$ as well. For every measurable $B \subseteq E$, we consequently have

$$u^* M_0(B) u = 0, \quad u^* M_1(B) u = 0.$$

The ordinary matrix $M_0(B)$ is positive, so $M_0(B)u = 0$ by Lemma 9.1. The Hahn matrix $M(B)$ is positive and has zero quadratic value at u , so $M(B)u = 0$ too. Subtracting yields $\varepsilon M_1(B)u = 0$, whence $\sigma_u(B) = 0$. Thus the vector measure vanishes on every subset of E , which means $|\sigma_u|(E) = 0$. This proves (36).

Sufficiency. Assume (i)–(iii), and fix an event E . Put $P = M_0(E)$ and $Q = M_1(E)$. The matrix P is positive. If an ordinary vector $u \in \ker P$, then $\mu_u(E) = 0$, and (ii) gives $\nu_u^-(E) = 0$. Hence

$$u^* Q u = \nu_u(E) = \nu_u^+(E) \geq 0.$$

The compression of Q to $\ker P$ is therefore positive. If in addition $u^* Q u = 0$, then $\nu_u^+(E) = 0$. The measure $\mu_u + \nu_u^+$ vanishes on E , so (iii) gives $|\sigma_u|(E) = 0$ and in particular $Q u = \sigma_u(E) = 0$. The exact hypotheses of Theorem 9.2 are now satisfied. It follows that $M(E) = P + \varepsilon Q$ is positive on every Hahn vector. Since E was arbitrary, M is positive. $\square$

The theorem is a null-ideal characterization, not a finite test in u . It quantifies over all ordinary vectors and all measurable null sets. Its content is that these ordinary measure-theoretic conditions are exactly sufficient to handle the vastly larger family of Hahn-valued vectors in (32).

Corollary 10.2 (Eventwise kernel and rank). *Under the equivalent conditions of Theorem 10.1, for every event E ,*

$$\ker M(E) = (\ker M_0(E) \cap \ker M_1(E)) \otimes_{\mathbb{C}} K_{\Gamma}.$$

If C_E is the compression of $M_1(E)$ to $\ker M_0(E)$, then

$$\text{rank } M(E) = \text{rank } M_0(E) + \text{rank } C_E.$$

Proof. Apply the kernel and rank clauses of Theorem 9.2 to each event separately. $\square$

This corollary is eventwise. It is not a claim of a common kernel for all events, nor of a spectral measure theorem for an infinite-dimensional Hahn Hilbert space.

10.3 Reduction to the known scalar criterion

Corollary 10.3 (Scalar specialization). *For finite real signed measures μ_0, μ_1 , the coefficientwise Hahn measure $\mu_0 + \varepsilon\mu_1$ is positive if and only if*

$$\mu_0 \geq 0, \quad \mu_1^- \ll \mu_0.$$

In dimension one, condition (36) adds no restriction.

Proof. For $u \neq 0$, $\mu_u = |u|^2 \mu_0$, $\nu_u^\pm = |u|^2 \mu_1^\pm$, and $|\sigma_u| = |u| |\mu_1|$. If $\mu_0 + \mu_1^+$ vanishes on an event, then μ_1^- also vanishes there by its absolute continuity with respect to μ_0 . Hence $|\mu_1|$ vanishes there, proving (iii). The case $u = 0$ is trivial. $\square$

This specialization is the scalar result already in the repository [9]: it is `herg:cor:twoscale`, the two-exponent case of `herg:thm:nullideal = meas:thm:nullideal`, and `FORMALIZATION.md` records a Lean proof of it over an arbitrary measurable space. The Herglotz report states it for regular Borel measures on a compact Hausdorff space; the proof above uses neither topology nor regularity. The new matrix ingredient is not scalarization itself, but the extra vector-variation condition that prevents a residual zero direction from coupling to a positive leading direction.

10.4 Examples showing the role of each condition

Example 10.4 (All scalar null tests pass, but matrix positivity fails). Let ρ be any nonzero finite positive measure and use the matrices P, Q from Example 9.3. Set $M_0 = P\rho$ and $M_1 = Q\rho$. For $u = (a, b)$,

$$\mu_u = |a|^2 \rho, \quad \nu_u = 2 \operatorname{Re}(\bar{a}b) \rho.$$

If $a \neq 0$, then $\nu_u^- \ll \mu_u$; if $a = 0$, then $\nu_u = 0$. Thus condition (ii) holds for every u . For $u = e_2$, however,

$$\mu_u = \nu_u^+ = 0, \quad \sigma_u = e_1 \rho \neq 0.$$

Condition (iii) fails. On any event of positive ρ -measure, the Hahn vector $(-\varepsilon, 1)$ detects a negative value. This works even on a one-point space, so the missing condition is not a subtlety of infinite measure spaces.

Example 10.5 (An indefinite coefficient is harmless). Use instead Q from Example 9.4 and again put $M_0 = P\rho$, $M_1 = Q\rho$. The measure is positive, because each event matrix is a nonnegative ordinary multiple of a positive definite Hahn matrix. For $u = (a, b)$ with $a \neq 0$, the leading control measure is already a positive multiple of ρ . For $a = 0$ and $b \neq 0$, $\nu_u^+ = |b|^2 \rho$ supplies the control that (iii) needs. Thus the vector condition permits the coupling, even though Q is indefinite as an ordinary matrix.

Example 10.6 (No bounded density is necessary). On $(0, 1]$ with Lebesgue measure, take

$$M_0 = I_d dx, \quad M_1 = -x^{-1/2} I_d dx.$$

Every entry has finite total variation. The density $x^{-1/2}$ is unbounded, but for each nonzero ordinary u , both the scalar negative part and the vector variation are absolutely continuous with respect to $\mu_u = \|u\|^2 dx$. Thus $M_0 + \varepsilon M_1$ is positive. Directly, if an event has positive measure, its matrix mass is

$$\left(|E| - \varepsilon \int_E x^{-1/2} dx \right) I_d > 0;$$

if it has zero measure, both coefficients vanish. The infinitesimal comparison is made separately for each event; no ordinary uniform bound on density ratios is needed.

11 What has been established, and what remains

11.1 Two distinct hierarchies of probes

The simplest coupling example $\begin{pmatrix} 1 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}$ has coefficients in a one-generator Hahn subfield. Ordinary complex vectors miss its negativity, but a finite-lattice vector detects it immediately. The prime-tail matrix is more severe: every finite-lattice vector, every vector algebraic over all monomials, and every vector in the valuation closure of the finite-lattice field misses it. A vector using the tail itself detects it immediately.

These are different levels of restriction. They should not be collapsed into a statement that “finite calculations cannot detect positivity.” In both examples an exact 2×2 determinant does detect the failure. What changes is the support needed in a *quadratic witness*, not the number of rows or the degree of a determinant.

11.2 Results and imported inputs

The proofs give a short dependency structure. Character extension gives the algebraic support-index bound. The support-index bound yields the exact primordial degree law and the algebraic closedness of the probe field. A fresh-prime coset argument gives the stronger simultaneous valuation separation. One congruence and a leading-term comparison then give maximal hidden negative inertia. The closure theorem is an approximation contradiction and does not require a new support lemma.

The matrix-measure theorem is logically independent of the prime-tail construction. Its only algebraic ingredient is the exact two-scale Schur-complement criterion, and its measure-theoretic ingredient is the ordinary characterization of a vector measure’s null sets by its total variation. The two parts meet conceptually in the difference between ordinary vector tests and all Hahn vector tests.

11.3 Questions genuinely left by this work

Question 11.1 (More than two measure scales). For a finite or transfinite well-ordered family of ordinary Hermitian matrix measures $\sum_{\gamma} t^{\gamma} M_{\gamma}$, is there an equally intrinsic null-ideal characterization that incorporates the Schur corrections generated by off-diagonal coefficients?

Theorem 10.1 answers this only when there are two scales. Example 9.5 explains why simply repeating its annihilation condition at each coefficient cannot be correct: later coefficients can compensate for earlier off-diagonal effects. No answer for arbitrary supports is claimed here.

Question 11.2 (Optimal probe classes under prescribed support closure). Given a support-family subfield $E \subseteq F_{\Gamma}$, which closure conditions on its supports imply order density, and which quantitative support invariants bound the degree of approximants to an omitted valuation ball?

Proposition 8.1 reduces qualitative quadratic testing to density. The prime-tail construction supplies one exact quantitative answer. It does not classify all support families.

Question 11.3 (Relative precision costs). How does the exact approximation law change when the ground field already contains an infinite prescribed collection of monomial roots, or when an algebraic approximant is charged by a tower of extensions rather than its total field degree?

The present theorem deliberately fixes the ground field $\mathbb{C}(t^{\delta})$. A general extension-degree complexity theory would require a separate cost model and should not be inferred from the primordial formula.

11.4 Non-claims

No result here gives a new construction of the surreal field, a new spectral theorem over real-closed fields, or a general solution to a published surreal conjecture. The algebraic independence theorem is ordinary algebraic independence, not differential independence for the Berarducci–Mantova derivation. No integration on the full proper-class surcomplex plane, no positive spectral measure representation, and no infinite-dimensional Hilbert completion are constructed.

The valuation topology used for closures is the topology of a fixed set-sized Hahn workspace. It is not the full fine topology on all of **No** or **No**[i]. The actual-surreal corollaries use normal-form transport, not a claim that countable partial sums converge in that fine topology.

Finally, the novelty assessment is a research judgment with limited search coverage. The written proofs can be checked independently of that assessment. A future discovery of a prior formulation would change the priority claim without changing the validity or usefulness of the explicit constructions.

Consolidated ledger. The source states its limitations where they arise. Each is kept in place; the list collects them, with the place where each stands. Items 1–24 are the source’s; items 25–27 were added when the report joined the collection.

1. Not refereed and not Lean-verified; no Lean source accompanies the report, and Appendix B gives only a possible formalization order (title page, Section 1, Appendix B).
2. No named published conjecture is claimed settled, and no general solution to a published surreal conjecture is given (title page, this subsection).
3. Priority is not certified. The novelty assessment is a research judgment from a pinned, targeted comparison; negative searches do not establish absence; the repository comparison is not a theorem-equivalence check; the Rayner paper and the Gonshor book were not audited line by line (Section 1, Appendix A).
4. Background, not contribution: the support-family construction of $\mathcal{P}_\Gamma$, diagonal character automorphisms and finite-index character arguments, finite Hermitian spectral algebra, the Hahn closedness theorem, the positive-support lemma, Schur complements and singular perturbation (abstract, Sections 2, 3 and 9).
5. That a nondense subfield can miss a negative quadratic value is not new, Proposition 8.1 is an elementary organizing fact, and nothing says that a Hahn field has no proper dense subfield (Sections 1 and 8).
6. “Hidden” refers only to vector probes from the named subfield. Determinants and principal minors detect A_r at once, and the construction is no obstruction to an exact positivity algorithm working in the field generated by the entries; there is no undecidability or general algorithmic-impossibility claim (Sections 1 and 6; the delivered README).
7. Strong summation is not valuation convergence of finite partial sums; closures use the valuation topology of one set-sized workspace, not the fine topology of **No** or **No**[i]; the surreal corollaries use normal-form transport, not fine convergence (Section 2, this subsection).
8. Proposition 2.3 concerns formal evaluation at positive-order arguments, not every version of surreal exponentiation (Section 2).
9. Lemma 3.2 does not assert that $[B : G]$ divides D .
10. The primordial law measures field degree over $\mathbb{C}(t^\delta)$ at the stated strict precision only: not expression length, running time, or a cost over an enlarged ground field; reaching a valuation without crossing it need not match the next coefficient (Section 4, Remark 4.5).
11. No novelty is claimed for the transcendence-degree cardinality in Corollary 5.3.
12. The independence is ordinary algebraic independence, not differential independence for the Berarducci–Mantova derivation (this subsection).
13. The sharpness statements concern dimension and inertia, not the order type of the supports of the

- entries; no support-order threshold is asserted (Section 6).
14. The matrices of Corollary 7.4 do not have all entries in the Levi–Civita field; the corollary does not say that a matrix defined over that field can be probe positive there and indefinite in an extension (Section 7).
 15. The surreal transport uses no set containing all surreal numbers and no class-indexed summation; the statement about coordinates algebraic over surreal monomials excludes coordinates whose defining parameters contain the prime tail (Section 8).
 16. Theorem 9.2 is not an arbitrary-support criterion in disguise, and Theorem 10.1 uses exactly two scales and coefficientwise countable additivity (strong Hahn summability of event masses is not assumed); the arbitrary-support matrix-measure problem is not settled (Sections 9–11).
 17. Theorem 10.1 is a null-ideal characterization, not a finite test in u (Section 10).
 18. Corollary 10.2 is eventwise: no common kernel for all events and no spectral measure theorem for an infinite-dimensional Hahn Hilbert space (Section 10).
 19. The scalar specialization, Corollary 10.3, is already in the repository (Section 10).
 20. No integration on the full proper-class surcomplex plane, no positive spectral measure representation, no infinite-dimensional Hilbert completion, no new construction of the surreal field, and no new spectral theorem over real-closed fields (this subsection).
 21. The prime-tail construction does not classify support families, and no extension-degree complexity theory is to be inferred from the primordial formula (Section 11).
 22. The finite checks are diagnostics: the exhaustive two-scale test covers only real matrices of sizes 1–3 with the stated entries, the stabilizer test does not check independence, and no finite test proves a statement about all measurable sets, ordinary vectors, Hahn supports or surreal workspaces (Appendix B).
 23. No result is claimed in the multiplier or small-divisor directions of the dynamics report, which the source considered and did not pursue (Appendix A; `source_audit.md`).
 24. The repository comparison is valid at its pin only; the repository has changed since (Appendix A; the changes that matter are in Appendix A.3).
 25. Theorem 10.1 answers no question posed in the Hahn–Herglotz report: the sentence it addresses is a non-claim there, and the matrix case with more than two exponents remains open (Question 11.1; Section 1.6).
 26. Theorem 5.2 neither implies nor follows from the independence theorems of the tail-spans and bounded-support reports; the comparison uses incomparable bases for $\Gamma = \mathbb{Q}$, with different witnesses (Section 1.6).
 27. The Lean proof of the scalar two-scale criterion recorded in `FORMALIZATION.md` covers no matrix statement and no prime-tail statement of this report (Section 1.6).

A Provenance, repository audit, and literature

A.1 Provenance of this report

This report is built from *one* manuscript: the archive `hidden_negative_directions`, manuscript 06 of batch 19 of the collection, dated 22 September 2026 and pinned to repository commit `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0`. It was placed in commit `52c7ab6` as a new report, because it continues themes and non-claims of existing reports but answers no question a report names. There was no second source; nothing was merged and nothing was selected away. Every result, proof, example, question and limitation of the manuscript is printed here.

The delivered verification program `code/verify.py`, build script `code/build.sh`, recorded outputs `data/verification.json` and `data/verification.txt`, the build summary `data/build_report.json` (which describes the *delivered* PDF, not this one), the dependency pin

`data/requirements.txt`, and the audit note `source_audit.md` are kept byte-identical. The audit note records the source’s own audit at its pin; corrections are made here, not in that file. The manuscript’s article and README were rewritten, and the delivered PDF was replaced by a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `hnd:`. The source’s 80 labels are kept, unchanged after the prefix, and nine were added (`hnd:sec:conventions`, `hnd:sec:position`, `hnd:sec:nonclaims`, `hnd:app:provenance`, `hnd:app:pinned`, `hnd:app:reproduce`, `hnd:rem:degreescopes`, `hnd:q:morescales`, `hnd:app:files`).
- (2) The valuation closure of $\mathcal{P}_\Gamma$, written $\mathcal{C}_\Gamma$ in the source, is $\overline{\mathcal{P}}_\Gamma$, to avoid the Hahn–Herglotz report’s $\mathcal{C}_\Gamma(X)$. The meaning of “positive” is fixed once, and restricted positivity is always qualified by its probe set (Section 1.5); the abstract now says so. Two source phrases were qualified accordingly: the non-claim after Corollary 7.4 and the count of positive semidefinite pairs in Appendix B.
- (3) Section 1.6 (relations to other reports) and the consolidated non-claims ledger in Section 11.4, with three items added, are new. Short pointers to labels of other reports were added where the source cited the repository generically.
- (4) Repository statements that were true at the pin but are no longer accurate or complete are corrected in Appendix A.3, keeping the pin as provenance.
- (5) Appendix B names the files as placed and records a rerun of the checks.

No mathematical statement was changed. Where the edit had to choose: the closure symbol (renamed, as above, rather than renaming the other report’s); whether to call Theorem 10.1 an answer to the Herglotz exclusion (it is called the two-scale case of an excluded criterion, since the exclusion is a non-claim, not a question); and the title, which is kept, with the directory named `hidden-negative-hermitian-directions` to say what kind of direction is hidden.

A.2 Pinned repository scope

The repository comparison uses

`048b72cf7cbfc8ab246e4f73788c10460cb3f6e0`

for <https://github.com/VladimirReshetnikov/Surreal>. The inspection was performed on 22 September 2026. The repository may have changed after that snapshot.

The reader map at this pin describes 36 reports and stresses that the collection consists of research drafts, with separate mathematical, finite-computational, and Lean-verification evidence. We inspected that map and the relevant spectral and Herglotz report guides, together with the dynamics guide to exclude another initially considered direction. We did not read every manuscript and every Lean file line by line.

Inspected item	Consequence for this manuscript
<code>docs/README.md</code>	Existing topics include finite and infinite spectral theory, Hahn measures, Herglotz positivity, tail-span differential transcendence, normal forms, and several analytic theories. These subjects are not presented here as missing wholesale.
<code>spectral-theory/README.md</code>	Hermitian diagonalization, finite congruence, Schur corrections, finite-index monomial extensions, and character calculations already occur. The present manuscript uses these ideas but claims neither their invention nor their absence from the repository.

Inspected item	Consequence for this manuscript
<code>hahn-herglotz-positivity/README.md</code>	Contains the scalar null-ideal criterion, two-scale scalar absolute-continuity condition, matrix halo normalization, and negative-mass examples. Explicitly excludes a matrix-valued null-ideal criterion. Theorem 10.1 supplies the two-scale matrix case, not an arbitrary-support replacement.
<code>dynamics-and-normal-forms/README.md</code>	Already contains extensive multiplier and surreal-frequency small-divisor results. No new result in those directions is claimed here.
Targeted code search	Searches for “algebraic probes” and “finite-lattice” returned no matches. Such negative searches are limited by terminology and indexing; they do not establish absence of an equivalent theorem.

The shortened report paths in the table are under `docs/surcomplex/`. The long report guides were not uniformly returned in full by the connector; the conclusions above are confined to the returned relevant portions. In particular, this was a targeted scope audit, not a repository-wide theorem-equivalence check.

A.3 Corrections since the pin

The subsection above is the source’s, as of its pin. Checked against the repository for this report:

- *Reader map.* At the pin `README.md` did describe 36 reports. Since then commit 52c7ab6 has placed four new reports, this one among them.
- *Negative searches.* “Algebraic probe” occurs nowhere under `docs/` at the pin, as the source found. “Finite-lattice” does occur at the pin, in an unrelated sense: `tate:cor:lattice` of `surcomplex/hahn-tate-uniformization` is “A finite-lattice degree formula” for torsion fields over K_Γ with $\Gamma \cong \mathbb{Z}$, and `FORMALIZATION.md` lists it; “finite lattice” also names `ent:mv:lem:lattice` of `surcomplex/entire-functions-at-arbitrary-rank`. Neither concerns probe fields or positivity, so the source’s conclusion stands, but its search result was incomplete.
- *Herglotz and measures.* The table entry for the Herglotz report is accurate; its exclusion is quoted in Section 1.6. The source did not mention that the transfinite scalar criterion is proved in `surreal/hahn-valued-measures-and-probability` (`meas:thm:nullideal`) and that the scalar two-scale criterion has a Lean proof recorded in `FORMALIZATION.md`; both are stated in Sections 1.6 and 10.
- *Tail-spans.* The source cites the reader map for the tail-span constructions. The precise relation, including the shared binary-prefix device, the incomparable bases and `tail:rem:indexfamily`, is in Section 1.6.
- *Generic citations.* The source’s citations of the reader map for Hahn closedness and for the orientation of normal forms are made specific: `polynomial:prop:workspace`, `found:sub:positivesupport` and `NOTATION.md`.
- *Same batch.* `surreal/transcendence-over-bounded-support`, placed with this report, could not have been seen by the source; its relation is in Section 1.6.

Between the pin and the placement commit, the spectral-theory, Hahn–Herglotz, tail-spans and measures reports received no change of substance to the material compared here.

A.4 Literature positioning

Kuhlmann and Serra’s work on Hahn automorphisms supplies the established context for diagonal character actions [5]. Krapp, Kuhlmann, and Serra study support families defining subfields [4], in the tradition of Rayner [3]. Kuhlmann’s work on dense subfields of henselian fields provides broader density context [6]. Gonshor supplies the classical surreal normal-form setting [1].

The automorphism paper was inspected in its arXiv PDF; the support-family and dense-subfield papers were inspected in available arXiv text. The original Rayner paper and the full Gonshor book were not audited line by line; they are cited for established background. Additional keyword searches did not locate the exact package of statements here, but some searches returned low-relevance results. Such searches are not evidence strong enough to certify priority.

The candidates for independent originality are the exact primordial precision theorem, the almost-disjoint-tail separation over the full finite-lattice field and its closure, their maximal-hidden-inertia application, and the vector-total-variation formulation of the matrix-measure criterion. The elementary density equivalence and the support-field construction are intentionally outside that claim.

B Finite verification and proof boundaries

The report directory contains `code/verify.py`. It uses exact integer, rational, and symbolic calculations; it does not substitute a small floating-point number for an infinitesimal. Its captured output is in `data/verification.json` and `data/verification.txt`. The recorded run passed all tests: 5946 matrix pairs, of which 1823 were positive semidefinite, and 35100 exact fresh-prime nonmembership comparisons. The script checks the following finite statements.

1. For $1 \leq r \leq 4$, the congruence identity $A_r = T^* \text{diag}(1, -\varepsilon^2 I_r) T$, determinant, and explicit negative vectors, using independent symbolic real h_j .
2. For $1 \leq N \leq 6$, the cyclic character stabilizer of the first N prime-tail exponents modulo P_N . It is trivial in each case, so the finite orbit has size P_N .
3. Fresh-prime coset separation for bounded denominators $1 \leq m \leq 60$ and primes at most 97, with exact rational arithmetic.
4. Every pair consisting of a diagonal 0–1 matrix P and a real symmetric matrix Q with entries in $\{-1, 0, 1\}$ in dimensions 1, 2, 3. There are $6 + 108 + 5832 = 5946$ pairs. It compares Theorem 9.2 with the signs of all principal minors of $P + tQ$, using their first nonzero polynomial coefficient.
5. The determinant identities in the forbidden-coupling, allowed-coupling, and three-scale repair examples.

The exhaustive test in item 4 covers only those finite real matrices. It does not establish the theorem for arbitrary complex matrices, arbitrary dimension, or arbitrary Hahn value groups. Item 2 checks finite stabilizers, not the infinite algebraic-independence theorem. No finite test here proves a claim about all measurable sets, all ordinary vectors, all Hahn supports, or all surreal workspaces. Those quantifiers are handled by the written proofs.

B.1 Files as placed

The delivered files keep their contents and were placed as follows.

- `code/verify.py`: the program above.
- `code/build.sh`: the source’s build script.
- `data/verification.json`, `data/verification.txt`: the source’s captured run (Python 3.13.5, SymPy 1.14.0).

- `data/build_report.json`: the source's summary of its own PDF build; it does not describe this text.
- `data/requirements.txt`: the pin `sympy==1.14.0`; the source's README placed it at the top level.

The program prints to standard output only and never rewrites the files in `data/`. The build script changes to its own directory before running `latexmk` or `pdflatex` on `article.tex`; as placed under `code/` it does not find the article. To use it, copy the report directory and put the script beside `article.tex` in the copy, or run `latexmk` directly as in the README.

B.2 Rerun for this report

On a copy of the directory, under Python 3.14.4 with SymPy 1.14.0, the program printed the same five PASS lines and the same counts as `data/verification.txt`; only the Python version string differs. Its `--json` output agrees with `data/verification.json` in every field except the recorded Python version.

B.3 A possible formalization order

A future Lean development could begin with the finite congruence and the two-scale algebra lemma, continue with the scalar/vector null-set measure equivalence, and then develop support-index and fresh-prime lemmas. The latter require a precise API for character extension, coefficientwise Hahn automorphisms, and finite generation on a rational line. Existing repository support, complexification, and spectral components should be reused rather than duplicated.

This is an implementation outline only. No Lean source or claim of machine-checked coverage for the new theorems accompanies this report.

References

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- [7] V. Reshetnikov, Surreal repository, *The surreal and surcomplex reports*, docs/README.md, snapshot 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0, accessed 22 September 2026. [Pinned repository reader map](#).

- [8] Surreal research collection, *Surcomplex Spectral Theory: Singular Values, Infinitesimal Rank, Multiscale Stability, Exact Determinantal Ramification, and Hankel Square-Class Profiles*, report guide, snapshot 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0. [Pinned spectral-theory report guide](#).
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